

Ritvik Singal

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EDUCATION

University of Illinois Urbana-Champaign
Bachelor of Science in Computer Engineering

Urbana, IL

Expected May 2028

- Relevant Courses: Computer Systems & Programming, Data Structures Analog Signal Processing, Operating Systems
- Planned Coursework (Spring 2026): System Verilog, FPGA Design, Algorithms & Computation

SKILLS

Programming: C, C++, Python, RISC-V, CUDA C, Embedded C, Assembly, MATLAB, Linux, Docker

Embedded Systems: FreeRTOS, Zephyr RTOS, ARM Cortex-M, STM32 HAL, I2C, SPI, CAN, UART, Arduino, ESP32

Design: Altium (PCB design/verification), KiCad, SolidWorks (3D CAD), High-Voltage Systems

Hardware Tools: Oscilloscope, Waveform Generator, isoADC, DC Load, Soldering Iron, Power Tools

WORK EXPERIENCE

Embedded Engineer Intern | KiCad, Zephyr RTOS, Nordic nRF54H20/nRF Connect SDK

Chicago, IL

Curafitt

Jun 2026 - Present

- Recreating a complete **104-pin carrier PCB** in **KiCad** around the **Ezurio BL54H20 module**, integrating SWD programming, RF support circuitry and precision clocking for a **dual-core 320 MHz BLE 5.4 platform**
- Designing a battery-powered subsystem around the **Nordic nPM1300 PMIC** to manage USB-C charging for a **600 mAh Li-Ion cell** while integrating **16 MB QSPI flash** for **OTA firmware storage** and future data logging
- Developing a **two-board wearable architecture** with dedicated **I²C, SPI** and **4+ GPIO interfaces**, supporting dual-core firmware development, **Zephyr-based IPC** and future **TinyML** and sensor-fusion workloads

EXTRACURRICULARS

Course Assistant | Oscilloscope, Waveform Generator

Urbana, IL

Introduction to Electronics, UIUC (ECE 110)

Jan 2026 – Present

- Assisting **100+ students** per week during lab and office hours with **circuit analysis**, breadboarding and debugging of **DC/AC electronics** experiments involving diodes, **BJTs, MOSFETs**, sensors and **PWM motor control systems**
- Guiding students through **13 weekly lab** modules using oscilloscopes, multimeters, function generator and power supplies while reinforcing **Kirchhoff's Laws, Thevenin/Norton** equivalents and nodal analysis

Battery Systems Embedded Engineer | Embedded C, isoADC, Testing/Validation

Urbana, IL

Illini Electric Motorsports FSAE

Aug 2025 – Present

- Created isoADC test plans to characterize current measurement accuracy across **8** polling rates and multiple **sampling configurations**, using a Rigol DC-Load, STM32 and **shunt resistor setup**
- Executed detailed noise and **gain analysis** at varying loads and polling frequencies (**128–400 Hz**) by collecting **2000+ samples** across **64** configurations to identify optimal isoADC current sensing parameters, improving overall **ADC accuracy** by **20%**
- Reduced **Interrupt Service Routine (ISR) runtime variability** by **15%** across **5 MCU load scenarios** by collecting and analyzing over **10,000 ADC samples** using Python while varying gain, current and sampling parameters to benchmark **BMS performance**

Embedded Research Intern | Ghidra, Assembly, Coldfire ISA, Jython

Urbana, IL

Security and Privacy Research at Illinois (SPRAI), UIUC

Sep 2025 – May 2026

- Reverse engineered **railway communication firmware** on the Coldfire architecture using Ghidra and **Jython scripts**, analyzing **50K+ lines of embedded C** and assembly while auto-extracting control flow and I/O logic
- Collaborated with a **4-member R&D** team to apply **reverse engineering techniques**, enhancing system reliability by **12%** and ensuring consistent communication integrity across railway signaling environments
- Deconstructed the **Golconda function** within railway communication firmware to map its control flow, uncovering **3+ potential buffer** overflow and **authentication vulnerabilities** in embedded communication protocols

PROJECTS

Custom Unix-like RISC-V Operating System Kernel | Embedded C, RISC-V Assembly, Sv39 Paging, QEMU

- Implemented a **FAT-based NGFS file system** driver and write-back block cache in C, supporting concurrent reads via **RW locks** and **reducing redundant 512B** backing-device fetches by **80%+ on repeated access patterns**
- Engineered a **Sv39 three-level paging virtual memory** subsystem and **ELF loader with lazy page allocation** and deep-copy fork semantics, isolating **39-bit user address spaces** across concurrent processes
- Constructed **15+ POSIX-style system calls** including fork(), pipe(), exec() and open(), driving a **preemptive timer-based scheduler** and a **multi-process shell** with I/O redirection, pipelining and background execution.